RIME: Raw mmWave I/Q Evidence Meets Visual Geometry for Grid-Level Indoor Wall-Structure Mapping

Zichen Xu

Kang Chiao International School Kunshan
kevinxuxuzichen@gmail.com

Abstract

Reliable indoor spatial mapping is the prerequisite for search and rescue tasks, and this is also the weakness of pure visual mapping. Wall surfaces with no texture and reflections from mirrors cause visual mapping to yield seemingly reasonable structures that are geometrically wrong; in addition, LiDAR emitting beams encounter signal scattering within smoky scnearios that camera visual sensing fails. The millimeter wave (mmWave) radar has wave length four orders of magnitude longer than them, able to penetrate through these obscurants, so it it's the only modality that can continuously collect signals under fire, building collapse, and ruins. However, existing mmWave radar processing adopts constant false alarm rate (CFAR) detection and Doppler gate that inhibit waves produced by indoor static walls before fusion.

This paper presents RIME (Radar-Image Map Evidence), which fuses the two modalities, mmWave radar and camera visual mapping, at the evidence level. By discretizing the scene into $0.4\,\mathrm{m}$ grids, mapping is transformed into binary classificaiton on the 220 candidate wall edges. Geometric information is provided by the monocular reconstruction model Depth Anything 3 (DA3); walls' existence, orientations, and range metrics informaton are directly extracted from raw ADC I/Q data corresponding to every edge. The mmWave detection utilizes empty room's recorded data to set false alarm rate without relying on object speeds and solves the ambiguity of eight orientations that monocular reconstruction cannot eliminate; the fusion weights passed through designs: they allow radars to rectify wall edges with relatively low confidence, but do not allow wall partitions to be generated solely by radars.

Experiments are based on the $4\,\mathrm{m} \times 4\,\mathrm{m}$ maze scene collected by AWR1843BOOST radar and DCA1000EVM: pure visual method detects 18 out of 21 wall partitions with a F_1 score of 0.923, pure radar method has a maximum F1 score of 0.310, and the fused map can completely detect every wall partition with no false positive. Furthermore, sending the empty room data into the whole detection pipeline for verification can get zero amount of occupancy grid, exhibiting successful false alarm control. Lastly, the map, with no additional densification or smoothing operations, is deployed on a Unitree Go2 quatruped to realize the navigation loop. From the planner's actual performance, vision map crosses through false negative wall edges for 340 times out of 3240 queries of shortest path, which does not exist in the fusion map.

Keywords—Indoor mapping, millimeter wave radar, raw I/Q evidence, radar-vision fusion, occupancy grid.

Contents

I Introduction **3**

II Related Work **4**
- II-A mmWave Radar Sensation and Monocular 3D Reconstruction Priors 4
- II-B Radar-Vision Fusion Representation 5
- II-C Occupancy Grid Mapping, Path Navigation, and Vision-Language Model Deployment 5

III Method **5**
- III-A Overall Methodology 5
- III-B Notation 7
- III-C Stage A: Radar Signal Formation 8
- III-D Stage B: Wall Existence Evidence Extraction 8
- III-E Stage C: Visual Geometric Reconstruction 9
- III-F Stage D: Frame Arbitration and Grid Edge Fusion 11
  - III-F.1 Frame Arbitration 11
  - III-F.2 Grid Edge Fusion 12
- III-G Stage E: Structural Regularization 12
- III-H Deployment on a Quadruped 13

IV Datasets **13**

V Experiments and Results **15**
- V-A Main Result 15
- V-B False Alarm Control 17
- V-C Radar Evidence Support and Regularization 18
- V-D Parameter Sensitivity 20

VI Conclusion **20**

I Introduction

On 26th November 2025, a bamboo scaffold burns at in Tai Po, Hong Kong, with fire surging upward and into the residential unit, causing 168 people died, which included a firefighter disoriented within smoke [1]. Similarly, in June 2007, a super store fire in Charleston, South Carolina killed 9 firefighters due to smoke inhalation and burns [2]. A fatal accident report shows that, within 17 cases and 23 deaths, smoke continuation exceeding duration of breathing apparatus and poor visibility causing disorientation are the main issues [3].

Such smoke fatality is the motive of sending robots for search and rescue tasks. The development of embodied AI allows quadruped robots to traverse rubbles and staircases that wheeled robots cannot pass through, enables vision-language model (VLM) to realize semantic deducing [4], and achieves the sensation-plan-execution cycle in real time. Nonetheless, these techniques are largely based on visible environments such as bright warehouses, offices mapped in advance, and clean laboratories [5]. Formulating paths and executions requires mapping, and mapping relies on sensation. Therefore, the robot becomes blind in smoke scenes without an applicable sensation system.

LiDAR has high accuracy and generates dense point clouds, while it emits beams of 905-1550 nm wavelengths [6]. Smoke and dust particulates scatter these light waves in the same physics mechanism as they scatter the light waves camera sensor relies on. The millimeter wave radar breaks this inability: the occluders are nearly transparent to the 77GHz mmWave radar because its wavelength is four orders of magnitude longer than LiDAR's [7–9]. Within the three sensing modalities, mmWave radar is the only one that receives effective signals within poor visibility, with low cost and small volume, but it has a poor angle resolution of 15 degrees, making it hard to reconstruct the entire geometric structure [10]. The visual monocular 3D reconstruction model, i.e., Depth Anything 3 (DA3) [11], captures detailed geometric information through image sequences, while it tends to miss flat surfaces with no textures, generate wrong mapping under specular reflection, and is susceptible to dust [12]. Instead of analyzing textures to generate depth map, mmWave radar measures the echo energy to precisely determine the existence and orientation of obstacles.

However, existing mmWave system processes radar signals through constant false alarm rate and Doppler gating, which are designed for vehicle motion detection, producing sparse point clouds [13, 14]. Static walls common at indoor scenes have Doppler (speed) close to zero and are inhibited before fusion, so they cannot be effectively mapped [15]. Therefore, the radar evidence is processed based on the I/Q raw data to prevent static walls being filtered. This paper presents RIME (Radar-Image Map Evidence), fusing radar and image evidence to execute binary classification, in which depth image provides geometric information configuration capture with relative depth and mmWave Radar provides walls' existence, orientations, and metric ranges of obstacles.

RIME models indoor mapping as a set of binary classification and decision instead of dense reconstruction problem. The $4\,\mathrm{m} \times 4\,\mathrm{m}$ scene is discretized into $0.4\,\mathrm{m}$ girds, with 220 candidate wall edges and each assigned with labels of "wall existence" or "wall inexistence." This modeling method enables the two modalities to query at the same granularity, which also matches the input granularity of path planner and requires no extra densification.

The two modalities execute decision tasks based on their respective physics characteristics. mmWave radar produces range-azimuth power map after performing range FFT and angle FFT to ADC I/Q signals. The static walls with echoes of zero Doppler are therefore maintained and not filtered. Angular contrast and empty room threshold are then combined with frame stability gate for confident evidence. The evidence is projected onto the BEV occupancy map, and mmWave's reflection characteristic is considered (front radar detecting horizontal wall edges and left radar detecting vertical wall edges). Vision modality uses principal component analysis on the DA3 point clouds to decompose, later implementing wall layer selection, Manhattan rectification, and normalization. For every edge, geometric and skeleton scores are computed.

Subsequently, the two groups of evidence source are aligned and fused. Since monocular reconstruction

cannot determine its frame orientation, eight combinations of rotations and mirroring are compared: the one that overlaps the most with radar frame is selected. Radar score, geometric score, and skeleton scores are fused with respective weights, and there is a single acceptance threshold for the final binary classification. The radar weight is delibarately set below the acceptance threshold so that radar evidence can save wall edges close to the threshold but cannot create a wall edge on itself. After structural regularization, valid wall structures are closed, and the output map is deployed on a Unitree Go2 quadruped's A * planner.

The research topic is fusing DA3-IQ representation forms for better mapping and navigation performance, and the contributions of this paper are as follows:

1. **Method**. Processes radar raw I/Q data through range-azimuth power map and false-alarm control and visual data through layer selection and Manhattan rectification for later grid edge fusion.
2. **System**. Builds a pipeline by feeding the mapping of TI radar and monocular reconstruction into the Unitree Go2 quadruped.
3. **Results**. Experiments demonstrate that, after fusion, F1 score is raised from 0.923 to 1.000 and that wrong length of navigation path is decreased by 10.5% when deployed on a quadruped robot.

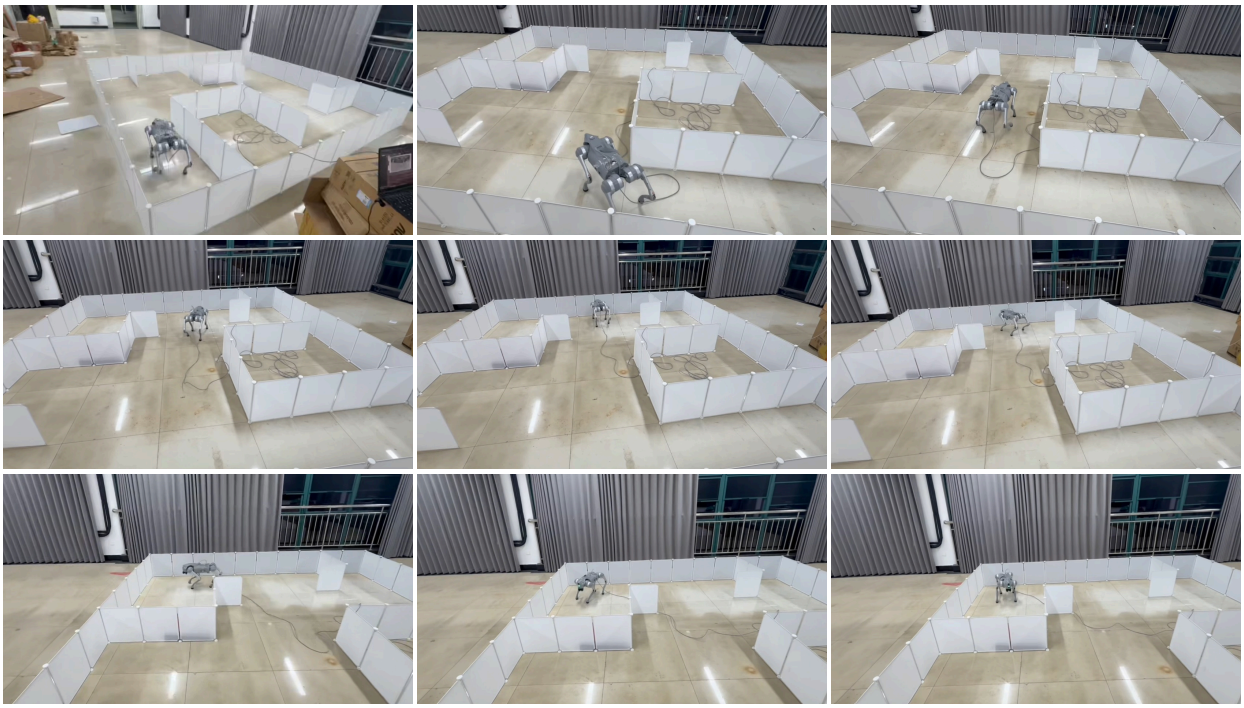


Figure 1: 3×3 sequenced images illustrating the process of entering maze and searching goal.

Here is the Github link for RIME Deployment Video Demonstration: https://github.com/kevinxuxuzichen-ship-it/RIME-Demonstration.

II Related Work

II-A mmWave Radar Sensation and Monocular 3D Reconstruction Priors

76-81 GHz FMCW radars have low costs and small volumes, still functioning under situations optical sensing like LiDAR fails and penetrating dust [7–9]. For detection, the typical chain of using range FFT, angle

estimation, CFAR processing, and Doppler gate outputs sparse point clouds for speedometer and SLAM algorithms and filters static objects common within indoor settings [13, 14, 16–21]. Its datasets provide detection points rather than I/Q raw data [22, 23]. For signal processing, more raw data are analyzed [24–26]. HawkEye extracts object outlines from heatmaps of low resolution [27], and milliEgo and BatMobility executes robotic ego-motion estimation under low visibility [28, 29].

Feed-forward deep learning models can reconstruct geometric scenes directly from unposed images. DUSt3R produces point clouds without labeled positions and orientations [30], MASt3R further expands dense features and metrics [31], VGGT outputs camera parameters, depths, and point tracks within same forward pass [32], and Depth Anything 3, based on the foundation of monocular depth models, predicts depth and ray maps [11, 33, 34]. These models are applied in mapping tasks [35–37], while textures and patterns may mislead predictions without outer sensors measuring information.

II-B Radar-Vision Fusion Representation

The core of fusion design is choosing the representation space. CenterFusion uses frustum projection to match detected points with the object proposals [38], CramNet infers the relationship between radar rays and echoes through cross attention. n mechanism [39], and CRN and RCBEVDet fuse after transforming image features into BEV space [40, 41]. 4D radar then expands into algorithms that evaluate autonomous driving datasets labeled with bounding boxes [42, 43]. These methods execute fusion after condensing raw signals into points, so the filtering effect still exists. Furthermore, the supervised systems' object proposals that center objects are not suitable for structures like wall partitions, and their feature concatenations lack audit trails for diagnosis.

II-C Occupancy Grid Mapping, Path Navigation, and Vision-Language Model Deployment

Occupancy grid mapping is still the standard representation form of indoor navigation [44]. SLAM system can output stable occupancy map with LiDAR or image inputs of high quality [45–47]. Researchers extract scene structures using estimation algorithms like RANSAC, Hough transform, and Manhattan world assumption [48–50]. A * algorithm is used for frontier exploration, collision prevention, and path optimization [51–54], which is adopted in hierarchical planning and optimal observing view selection [55, 56].

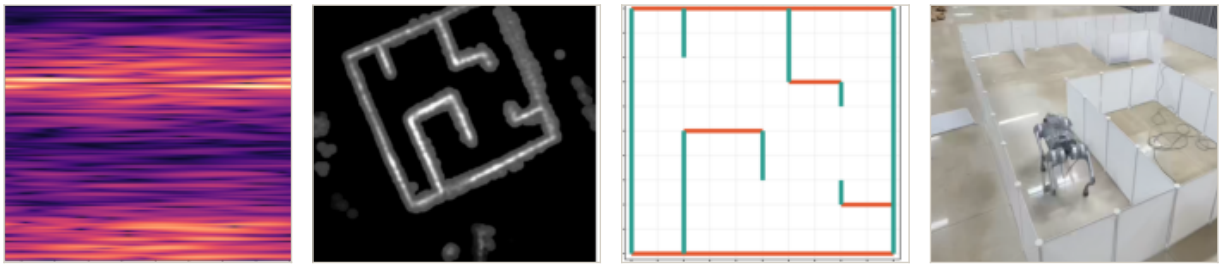
Pretrained VLM enables the potential of navigation in new, unseen environments [57]. CLIP on Wheels dismantles object navigation task into localization and environment detection [58]; LM-Nav combines language, vision, and actions [59]; VLMaps backprojects image features to generate BEV maps can be queried [60]; L3MVN and ESC utilizes information of common sense to select exploration frontier [61, 62]; lastly, VLFM adopts semantic associations to score frontiers on the value map [63]. Most VLMs are tested in simulation environment [64], and run on the cloud when deployed [65, 66]. In real scenes, VLMs have hallucination errors requiring other sensors to regulate and deducing latency.

III Method

III-A Overall Methodology

The mapping system consists of five stages (shown in Fig. 3):

- **Stage A (construction of radar signal)** aligns raw I/Q data with the virtual arrays, performs range FFT and spatial FFT, and generates range-azimuth power map for each view.
- **Stage B (extraction of wall existence evidence from power map)** removes the background noise that is dependent on range through angular contrast, uses empty room's recorded data to set the false



(a) what the radar returns (b) what vision proposes (c) what RIME decides (d) what it is for

Figure 2: From raw I/Q to a traversable map. **(a)** A range–azimuth power map from one radar, the sensor's actual output: the dominant structure is banding dependent on range, and no wall is individually distinguishable. **(b)** The wall-bearing layer of the monocular reconstruction, which resolves geometry but arrives unscaled, rotated by an unknown multiple of $90°$, and surrounded by returns off structure. **(c)** The fused decision on the $0.4\,\mathrm{m}$ edge lattice, matching the surveyed layout exactly on the 21 internal partitions. **(d)** The quadruped executing a path planned directly on (c). Neither modality suffices alone: radar reaches $F_1 = 0.310$, vision 0.923, the two together 1.000.

alarm threshold, maintains the evidence through stability, projects evidence onto BEV occupancy map, and applies directional constraints on front and left radars.

- **Stage C (reconstruction of visual geometric information)** divides the layers within DA3 point clouds, selects the wall layer, performs Hough transform and Manhattan rectification to align with grid axes, and computes geometric and skeleton scores for each edge.
- **Stage D (frame arbitration and the fusion of two modalities)** compares the overlap between eight visual combinations and radar occupancy map to resolve orientation ambiguity, uses reciprocal gate to filter radar evidence lacking visual suupport, and fuses three terms for the fusion score on every edge.
- **Stage E (regularization for closed wall structure)** relies on rules like boundary assertion, continuous length, and endpoint attachment to complete rectification.

Section III-H deploys the Stage E output on a quadruped for feasibility demonstration, focusing on the override of VLM decision over A * planning and labeling dynamic obstacles.

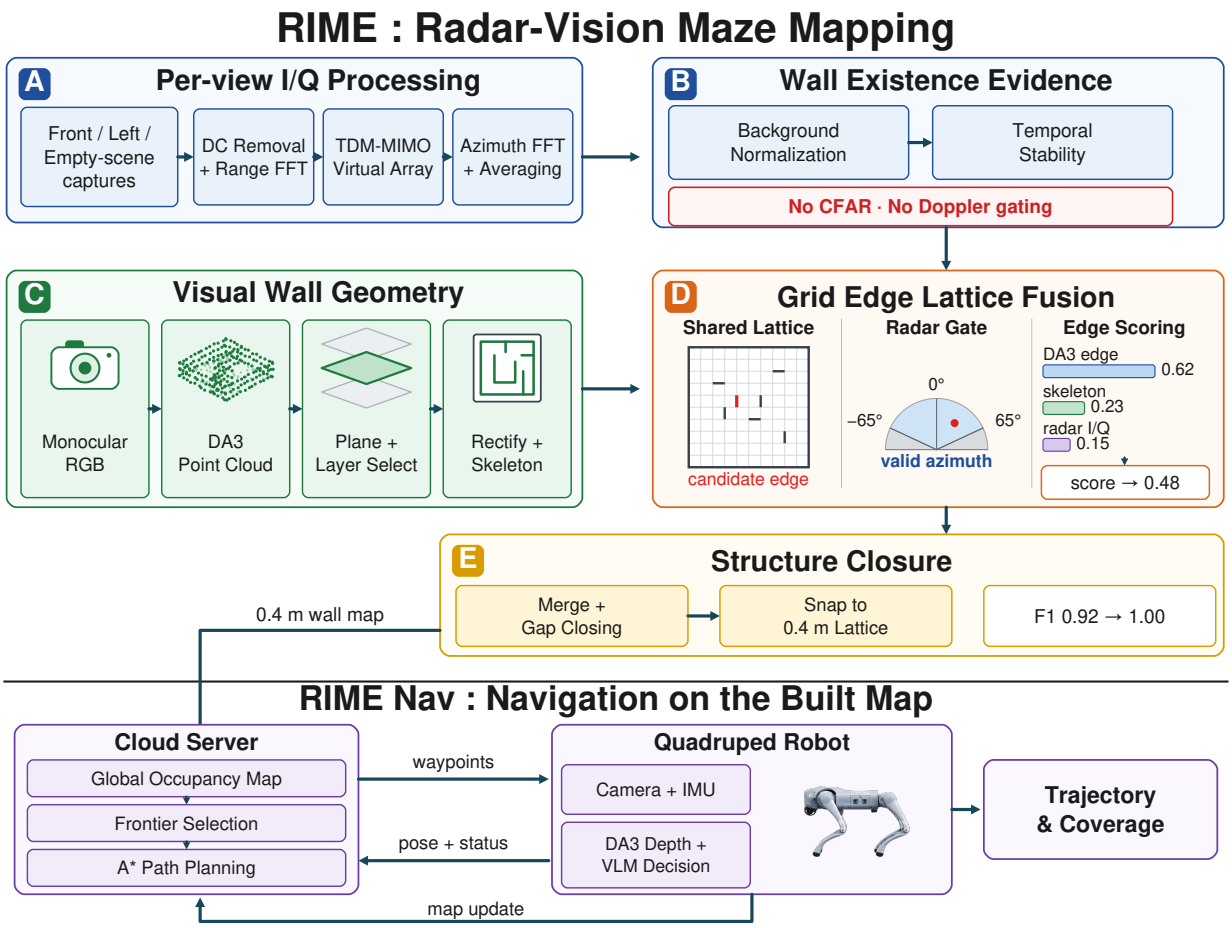


Figure 3: RIME system pipeline. Stage A-B uses I/Q data for wall evidence, stage C extract wall geometry from visual point clouds, stage D fuses radar and vision modalities, stage E rectifies the structure, and the output map is analyzed by RIME NAV for path navigation.

III-B Notation

The scene is a square of $4\,\mathrm{m}$ side length divided into 10×10 cells with each edge of $0.4\,\mathrm{m}$. There are 110 horizontal edges and 110 vertical edges, of which 40 are boundary edges and 180 are internal edges; there are 21 edges carrying a wall within the internal edges. Horizontal edges are indexed $e^H_{r,c}$ with $r \in \{0, \ldots, N\}$ and $c \in \{0, \ldots, N-1\}$, and vertical edges are indexed $e^V_{r,c}$ with $r \in \{0, \ldots, N-1\}$ and $c \in \{0, \ldots, N\}$. Table 1 includes the remaining symbols:

Table 1: Key symbols used across more than one stage. Symbols appearing only within a single equation are defined where that equation is introduced.

Symbol	Meaning	Value
$\rho,\ k$	range-bin index, azimuth-bin index	$< 256, < 256$
$P_v[\rho, k]$	range-azimuth power of view v	—
$C_v[\rho, k]$	angular contrast	—
τ	detection threshold from the empty scene	$8.13\,\mathrm{dB}$
$\mathcal{D}_{3\times3}$	3×3 binary dilation	—
$\mathcal{O}_v$	occupied cells detected in view v	$11, 8$

Table 1, continued

Symbol	Meaning	Value
i^H, i^V	radar edge votes	binary
$\mathcal{L}_{\text{main}}, \mathcal{L}_{\text{sec}}$	candidate point cloud layers	22,377, 5,437 pts
$\mathcal{M}$	normalized wall mask	400×400
g^H, k^H	geometry score, skeleton score	—
g_0, k_0^H, k_0^V	saturation constants	0.65; 18, 24
w_g, w_k, w_i	fusion weights	0.65, 0.25, 0.10
s^H, s^V	fused edge score	—
η	acceptance threshold	0.48
$m_a,\ J(a)$	wall strength and vote score of variant a	Fig. 6

III-C Stage A: Radar Signal Formation

Every view v gets F_v frames of 64 chirps $\times$ 3 transmitters $\times$ 4 receivers $\times$ 256 raw ADC complex samples in the form of $X = X^{\mathrm{I}} + \jmath X^{\mathrm{Q}}$. After subtracting ADC offset to prevent peak at zero range, Hann window w^{r} to prevent strong walls masking weak ones, and doing range FFT and angle FFT to convert beat frequencies into ranges and angles, the receive sequence for Tx_1 is reversed to align with opposite directions of transmitters [8]. The azimuth response of one single chirp is calculated as follows:

$$A_v[l, \rho, k] = \sum_{q=0}^{Q-1} w^{\mathrm{a}}[q]\, Y_v[l, q, \rho]\, e^{-\jmath 2\pi qk/K}, \tag{1}$$

where v indexes the three recordings, F_v is the number of valid frames in view v, l and q index the $L = 64$ chirps and the $Q = 8$ virtual channels, Y_v is the virtual array signal after range compression, w^{a} is a Hann window applied along the array, $K = 256$ is the number of azimuth bins, and $A_v[l, \rho, k]$ is the azimuth response of one single chirp. The range resolution is $\Delta\rho = 4.37\,\mathrm{cm}$, so the metric range of bin ρ is $\rho\Delta\rho$, and the azimuth angle of bin k is denoted θ_k. The power map is then obtained by averaging across chirps:

$$P_v[\rho, k] \;=\; \frac{1}{L} \sum_{l=0}^{L-1} \big|A_v[l, \rho, k]\big|^2 \qquad \text{—not} \quad \Big|\tfrac{1}{L} \textstyle\sum_l A_v[l, \rho, k]\Big|^2. \tag{2}$$

Eq. (2) calculates the average power at range bin at index ρ and angle bin at index k within L chirps. For moving objects, the phase drift carries Doppler information, while it becomes a distraction for static walls. Averaging the complex spectra cancels waves with each other due to phase drift, so taking square before averaging can prevent this and calculate the power out [14].

III-D Stage B: Wall Existence Evidence Extraction

Subtracting antenna leakage and noise background from P_v can remove power distractions above while maintaining angular peaks. Echoes reflected from objects (walls) at different distances possess different powers because the further distance wave travels dissipates more energy, and this is why subtraction is compared against angles at the same distance. The angular contrast is defined as

$$C_v[\rho, k] = 10 \log_{10} P_v[\rho, k] - \operatorname*{med}_{k'}\ 10 \log_{10} P_v[\rho, k']. \tag{3}$$

Based on the empty room recording, which represents the noise background, a threshold of q=99.7% is set on empty scene's angular contrast for later false alarm control and filtering. The detection threshold is set from the empty scene as

$$\tau = \mathcal{Q}_q\big(\{\, C_\varnothing[\rho, k] \;:\; (\rho, k) \in \mathcal{R} \,\}\big). \tag{4}$$

Eq. (4) takes the role of CFAR [13], but it does not incorporate speed so that static walls are not penalized for zero Doppler speed.

The median angular contrast $\bar{C}_v[\rho, k]$ (power intensity) across frames and stability $\sigma_v[\rho, k]$, defined as fraction of F_v frames that have $\bar{C}[\rho, k]_v \geq \tau$, ensure walls detected need to satisfy power intensity and stability at the same time. After polar bins are projected onto BEV cells through $x = \rho\Delta\rho \sin\theta_k$ and $y = \rho\Delta\rho \cos\theta_k$, $\bar{C}_v[\rho, k]$ and $\sigma_v[\rho, k]$ are max-pooled into $S_v[b]$ and $\Sigma_v[b]$. The occupancy map is obtained by

$$\mathcal{O}_v = \Gamma_{\kappa_{\min}}\Big(\{\, b : S_v[b] \geq \tau \ \wedge \ \Sigma_v[b] \geq \sigma_{\min} \,\}\Big). \tag{5}$$

With left radar placed at left side center and front radar placed at front side center both with a distance of $0.5\,\mathrm{m}$ to the boundaries, the coordinates are transformed to fit the radars' positions. Since wall surface only reflects energy wave perpendicular to it, directional constraints are necessary (Fig. 4), with left radar detecting vertical walls and front radar detecting horizontal walls. The radar edge votes (wall occupancy) with directional constraints are dilated to tolerate one cell of error:

$$i^H \leftarrow \mathcal{D}_{3\times3}(\mathcal{O}_{\mathrm{f}}), \qquad i^V \leftarrow \mathcal{D}_{3\times3}(\mathcal{O}_{\ell}). \tag{6}$$

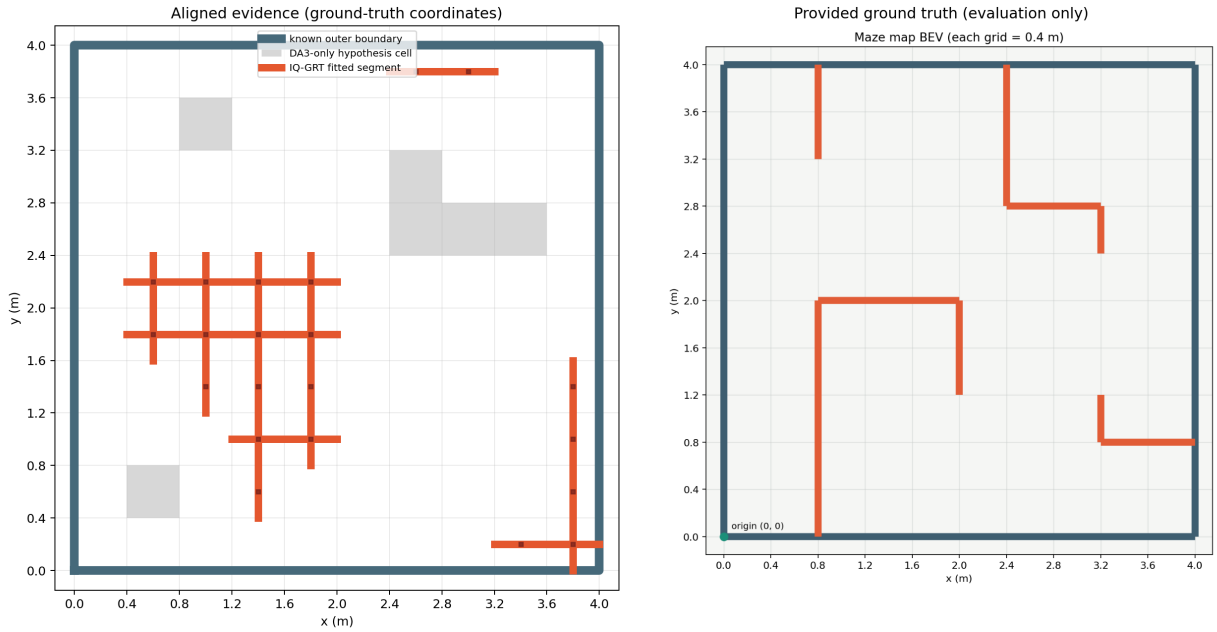


Figure 4: Left: wall segments cross with each other without directional constraints, demonstrating the necessity of directional constraints. Right: provided ground truth map.

III-E Stage C: Visual Geometric Reconstruction

For the visual point clouds of $\mathcal{P} = \{p_i\}_{i=1}^{37172}$, we take the median as center and perform PCA decomposition to get vectors $\mathbf{v}_1$ and $\mathbf{v}_2$ forming a plane and $\mathbf{v}_3$ along the plane normal [48]. The distance from a point to the plane is n_i, and two planes are divided along the n direction. The point cloud is divided along the normal direction into

$$\mathcal{L}_{\text{main}} = \{0.02 \leq n_i \leq 0.17\}, \ \mathcal{L}_{\text{sec}} = \{-0.42 \leq n_i \leq -0.20\}. \tag{7}$$

Through Canny and Hough Transformation [49], we select the layer with most abundance of lines aligning with the axis directions to find the wall structure layer:

$$\Lambda(\mathcal{L}) = \sum_{s \in \text{Hough}(I_{\mathcal{L}})} \|s\| \cdot \mathbb{1}\big[\angle s \text{ within } 10^\circ \text{ of an axis}\big], \tag{8}$$

where $I_{\mathcal{L}}$ is the raster image of layer $\mathcal{L}$ and $\|s\|$ is the length of a detected segment s. As a result, $\Lambda(\mathcal{L}_{\text{main}}) = 1717$ is 7.7 times of $\Lambda(\mathcal{L}_{\text{sec}}) = 222$, but the wall structure does not appear on the main plane from the graph, causing a misled conclusion.

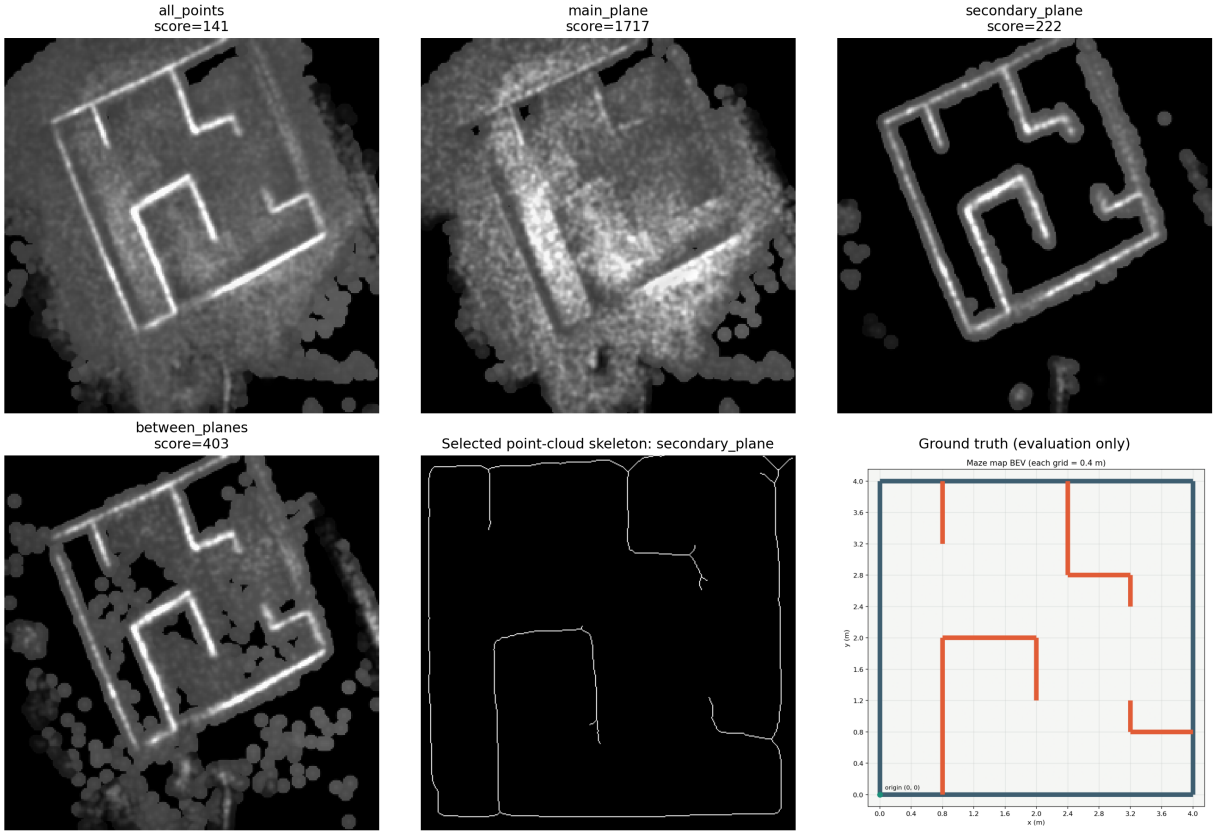


Figure 5: This wrong conclusion can be attributed to much higher density of points on the main plane compared to the secondary plane (22377 versus 5437), offsetting the effect of axis alignment. The secondary layer is selected for wall evidence.

Visually (Fig. 5), the secondary plane contains wall layer, while the main plane mainly contains floor layer. Therefore, $\mathcal{L}_{\text{sec}}$ is selected for the wall structure. In order to orient the segments from secondary layer into horizontal and vertical lines, Manhattan rectification is used to find the rotation angle [50]. The Manhattan angle is found as

$$\theta^\star = \arg\max_{\theta \in [0, \frac{\pi}{2})} \sum_j \ell_j \cos^2\big(2(\phi_j - \theta)\big), \tag{9}$$

where ϕ_j and ℓ_j are the orientation and the length of segment j, and $\theta^\star$ is the resulting Manhattan angle. The rotation angle resulting in maximum value represents a coordinate most aligned with segments. $\cos^2(2\cdot)$ ensures that segments with angles θ and $\theta + \pi/2$ (horizontal and vertical ones relative to the axis) earn the same score. For the Manhattan frame here, $\theta^\star = 34.76^\circ$ from 312 segments, with layer rotated by $-\theta^\star$.

After rectification, the secondary layer is binarized by an Otsu's threshold of 0.35, closed, maintained in largest connected region, and projected onto a 400×400 lattice pixel map through a projective matrix $\mathbf{H}$, obtaining $\mathcal{M} \in \{0,1\}^{400\times400}$. Bands having half width of $b = 7$ pixels ($\approx 7\,\mathrm{cm}$) center on each lattice line, and the geometric score is computed as

$$g^H[r,c] = \frac{1}{|\mathcal{B}^H_{r,c}|} \sum_{(u,v)\in\mathcal{B}^H_{r,c}} \mathcal{M}[u,v], \tag{10}$$

where $\mathcal{B}^H_{r,c}$ is the band of half width b centered on the horizontal edge at row r and column c. After $\mathcal{M}$ is skeletonized into center lines, the skeleton score is calculated by the sum of center line pixel votes of each nearest lattice line when they are $\leq 0.11\,\mathrm{m}$ close. The skeleton score is computed as

$$k^H[r,c] = \sum_{(x,y)\in\mathcal{K}} \mathbb{1}\big[\,\pi^H_{r,c}(x,y)\,\big], \tag{11}$$

where $\mathcal{K}$ is the skeletonized center line set, d_x and d_y are the distances from a pixel to the nearest vertical and horizontal lattice lines, and $\pi^H_{r,c}$ is the predicate that admits a vote. When $d_y = d_x$, add the vote to horizontal lattice line to prevent repeated counts.

III-F Stage D: Frame Arbitration and Grid Edge Fusion

III-F.1 Frame Arbitration

Principal component analysis can confirm the plane, while it cannot determine the orientation and mirror that align with radar mapping before fusion. Therefore, to find the correct orientation and mirror out of the eight possible combinations, the focus should be on the combination that overlaps the most with the radar evidence. Each variant is scored by its overlap with the radar detections:

$$\begin{aligned} J(a) = &\frac{\sum_{b\,\in\,\mathcal{D}_{3\times3}(\mathcal{O})} m_a[b]}{\sum_b m_a[b]} \\ &+ \frac{1}{4}\cdot\frac{\big|\{b : m_a[b] \geq 0.16\} \cap \mathcal{D}_{3\times3}(\mathcal{O})\big|}{\big|\{b : m_a[b] \geq 0.16\}\big|}, \end{aligned} \tag{12}$$

where $\mathcal{A}$ is the set of eight rotation and mirror variants and $m_a[b]$ is the wall strength of variant a in cell b, obtained by resampling the rendered reconstruction onto the 10×10 cell grid. The outer ring is excluded so that the asserted boundary cannot contribute to the score.

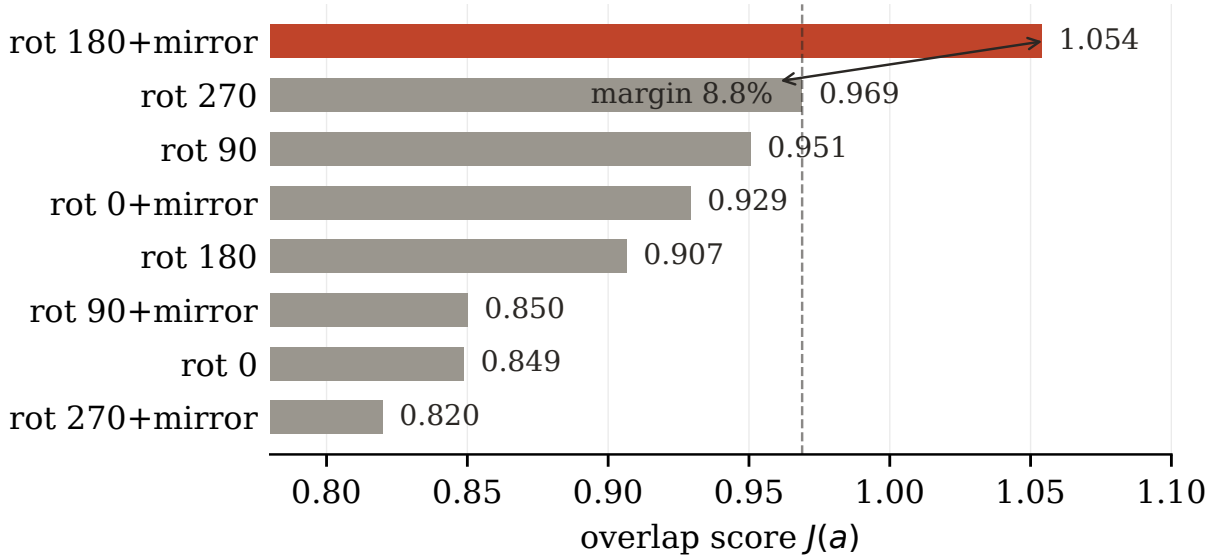


Figure 6: The eight combinations are scored based on Eq.(12). The combination overlapping the most is scored on a margin of 8.8% higher than the second one. Choosing a wrong combination may invert or rotate the map in other directions before fusion.

Using radar and vision modalities to constrain each other can ensure walls radar detect have to fall beside visual grids with confident wall strengths before fusion. The reciprocal gate is applied as

$$\mathcal{O}_v \leftarrow \mathcal{O}_v \cap \mathcal{D}_{3\times 3}\big(\{b : m[b] \geq 0.16\}\big). \tag{13}$$

III-F.2 Grid Edge Fusion

The three scores are saturating scaled to [0,1] and combined to calculate the final fusion score. The fused score of a horizontal edge is defined as

$$\begin{aligned} s^H[r,c] \;&= w_g \min\Big(\tfrac{g^H[r,c]}{g_0},\, 1\Big) + w_k \min\Big(\tfrac{k^H[r,c]}{k_0^H},\, 1\Big) \\ &+\; w_i\, i^H[r,c]. \end{aligned} \tag{14}$$

This works the same in vertical edges. Edges have to pass the threshold of $\eta = 0.48$ to be categorized as walls $\{e : s[e] \geq \eta\}$, with$(w_g, w_k, w_i) = (0.65, 0.25, 0.10)$, $g_0 = 0.65$, and $(k_0^H, k_0^V) = (18, 24)$.

Since $w_i = 0.10 < \eta = 0.48$, radar evidence cannot create a wall without the support of vision evidence. Radar evidence can save an edge to become wall when the sum of two visual terms lies between 0.38 and 0.48. Radar can therefore only change a decision when

$$\eta - w_i \;\leq\; s_{\text{vis}}[e] \;<\; \eta, \qquad \text{i.e.} \quad s_{\text{vis}}[e] \in [0.38, 0.48). \tag{15}$$

III-G Stage E: Structural Regularization

The fused occupancy map is passed through regularization rules for reasonable geometry [44]. The regularization operator $\mathcal{C}$ applies four rules:

1. The boundary edges are occupied (Frame Assertion).

2. Internal walls shorter than 2 cells are deleted.

3. Horizontal wall of length ≥ 2 cells can recover adjacent vertical wall at lower threshold of $s^V \geq 0.09$.

4. Partially detected long walls are completed to the boundaries, and vertical walls having lengths of 3 cells that connect far and float near are deleted due to snapping effect.

III-H Deployment on a Quadruped

The deployment feeds the map output from stage E into navigation module without adding densifying, smoothing, and interpolating. The rasterized map $\mathcal{G}_{\text{stat}}$ has a resolution of $0.04\,\text{m}$ per grid, and the walls are dilated into ellipse units with radius of $0.25\,\text{m}$ to leave space for safe quadruped width while navigation. The path is computed by A* algorithm using Euclidean heuristic function of distance [51], with unit and $\sqrt{2}$ costs on the 8 adjacent grids and small penalty of $\beta = 0.03$ per leftward step to break the tie.

The VLM is queried each step, and responses below confidence level of 0.5 are discarded. VLM only overrides the planner when the planner has no path or two agree within 90°:

$$\text{override} \iff \big(\Pi = \emptyset\big) \ \vee \ \big|\text{wrap}(\psi_{\text{vlm}} - \psi_{\text{A}^*})\big| < \tfrac{\pi}{2}. \tag{16}$$

Objects are identified as dynamic obstacles when the response has confidence level ≥ 0.45 and labeled dynamic in mobility or belong to $\mathcal{T}_{\text{dyn}} = \{\textsf{person}, \textsf{box}, \textsf{chair}, \textsf{object}\}$.

IV Datasets

RIME lacks public benchmark dataset for comparison because existing mmWave datasets provide detection results and scans of images instead of ADC raw data. Such reduction in dimensions is what this paper avoids. Therefore, no existing I/Q signal datasets at indoor wall edge level can be compared. RIME does not need training due to its framework. All thresholds and weights in the method section III are fixed, not learned, so no loss function and sets for training and testing are needed.

The two radars detect in orthogonal directions and remain static (Fig. 7). Both radars are set at the centers of side lines, with a distance of $0.5\,\text{m}$ from them. The region of interest (ROI) for radar detection is $|\theta_k| \leq 60^\circ$ because the arcsine mapping of the azimuth axis becomes unreliable near $\pm 90^\circ$. The wall partitions are made in white plastic material, which aligns with the goal of multimodal fusion: flat surfaces with no texture and mirror reflection effect. These surfaces, used for separate dataset collections for monocular reconstruction's camera sequence and radar data, can mislead visual feature extraction but produce relatively strong millimeter wave echoes.

The data are collected within experiments from TI AWR1843Boost, DCA1000EVM, and camera with Depth Anything V3 model [11] (Fig. 8), RIME system does not require training, and there is no public benchmark regarding indoor wall structure from raw I/Q [22, 23].

There are 21 internal partitions of walls, and the ground truth reference is not leaked during mapping. The radar operates with 76–81 GHz band and range resolution of $4.369\,\text{cm}$.

Table 2: Radar recordings.

Recording	Role	Frames	Used by
`...02.13.10-front`	scene, front view	28	Stage A–B
`...02.58.07-left`	scene, left view	28	Stage A–B
`...03.05.11-empty`	background model	29	Eq. (4)

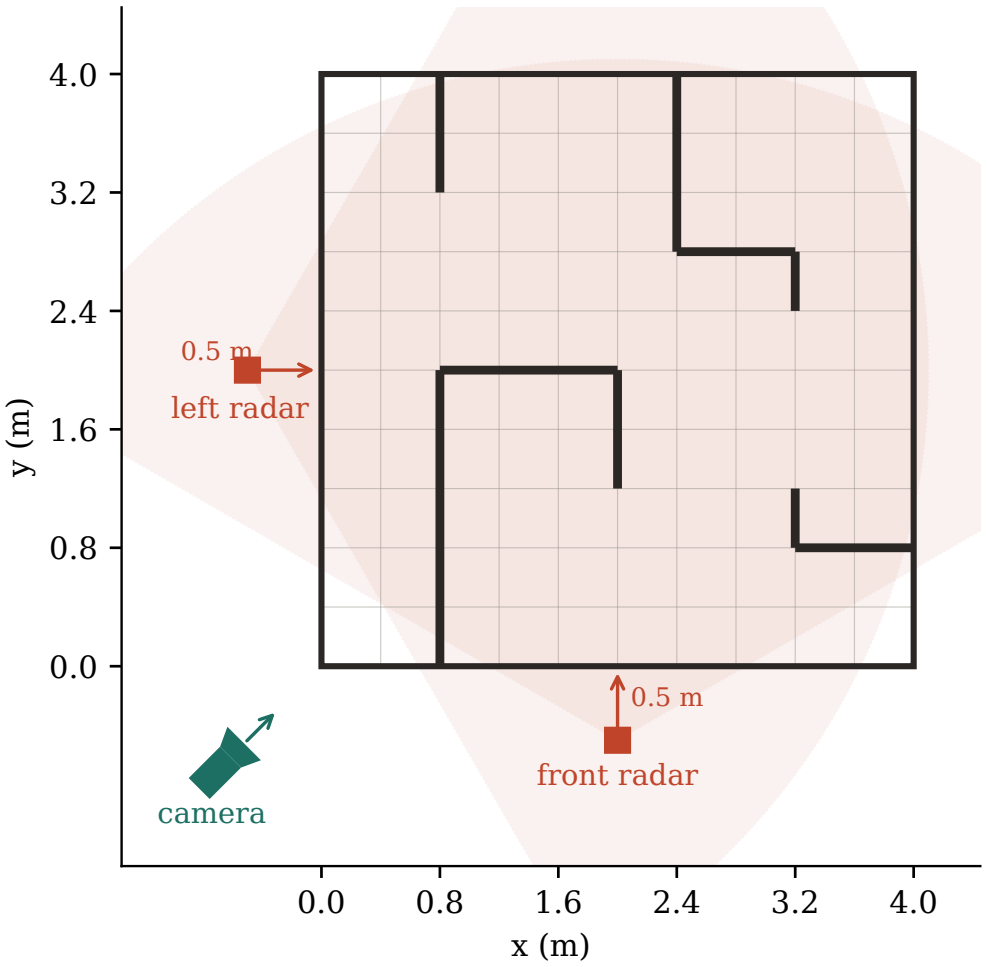


Figure 7: Configuration of experiment, with region of interest of $|\theta| \leq 60°$.



(a) radar rig (b) scene, front view (c) scene, left view (d) empty room

Figure 8: Hardware and map setup.

For feasibility demonstration, there are three trials of deployment using a Unitree Go2 quadruped and Qwen3.5-4B VLM.

Table 3: Path steps and VLM responses within deployment trials.

Trial	Steps	Path (m)	Fwd/L/R	Responses	Overrides
1	21	2.17	12/2/7	3	1
2	26	2.39	12/5/9	5	1
3	14	1.88	10/0/4	2	0
Total	61	6.44	34/7/20	10	2

V Experiments and Results

V-A Main Result

There is no external baseline because no prior methods possess the inputs and outputs of this radar-camera system [24,27,38,40]. Table 4 shows the geometric confidence and high precision of visual modality, and it also demonstrates how combining visual modality with radar modality closes the gap, raising F_1 score from 0.923 to 1.000.

Table 4: The main result on the 180 internal candidate edges, including 21 occupied edges, and the single variable ablation on radar factor ("Vision only" and "Both" all passed through stage E).

Configuration	Sel.	TP	FP	FN	P	R	F_1
Radar only, quantile operating point	29	4	25	17	0.138	0.190	0.160
Radar only, tuned threshold	30	7	23	14	0.233	0.333	0.275
Radar only, best pool	50	11	39	10	0.220	0.524	**0.310**
Vision only	18	18	0	3	1.000	0.857	0.923
Both (proposed)	21	21	0	0	1.000	1.000	**1.000**

Three conclusions can be made from table 4. First, under all test cases, solely relying on mmWave radar cannot reconstrcut the scene structure. If prioritizing precision, radar can only detect 4 out of 21 wall partitions; when prioritizing recall, althrough 20 actual wall partitions can be detected, 39 false positives are produced, of which the best F_1 is merely 0.310. This is not the problem setting threshold produces, but the problem of its intrinsic limited resolution: the radar of single chip has range resolution at centimeter level but azimuth resolution at 14.5 degrees.

Second, the vision side has opposite direction. Its precision rate achieves ideal 1.000, meaning every wall edge visually detected is real. On the other hand, its recall rate is 0.857, and the visual system does not give warnings or observe the mistake for the three missed wall edges, showing its confidence in geometric structure but inability to detect omissions.

Third, these two modalities are complementary to each other instead of information redundancy. After introducing radar evidence, three missed wall edges are recovered while maintaining the precision at 1.000. Therefore, the overall performance improvement can be entirely attributed to the increase in recall rate and does not bring false wall edges.

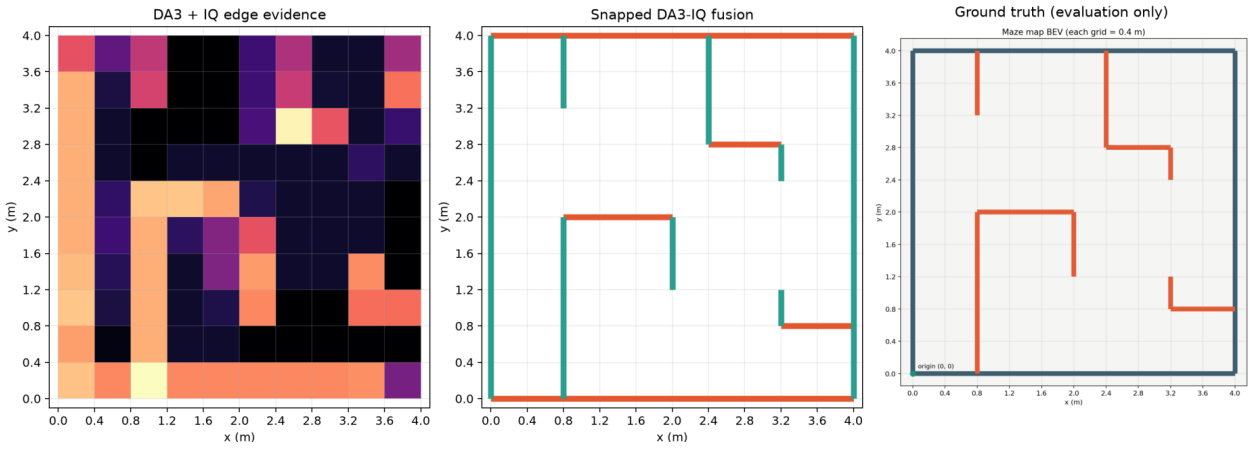


Figure 9: From per-cell evidence to a decided map.

The leftmost image shows evidence fusion at $0.4\,\mathrm{m}$ grid withoout applying threshold, with luminosity showing confidence level. The middle image shows the effective edges produced after stage E regularization, with orange representing horizontal edges and green representing the vertical ones. The rightmost image shows the surveyed ground truth results. Comparison shows that the fusion map has same prediction results with the ground truth map on all of the 21 wall edges.

Fused mapping decreases "wrong length" during navigation, which counts pairs with shortest path differing from the true one. On the vision only map, 340 out of 3240 pairs return a shorter path than the true one because the path planner formulates crosses through wall partitions that are not mapped but actually exist. Fused map minimizes such "wrong length".

Table 5: Path planning agreement with the surveyed map.

Map	Pairs	Reachability	Wrong length	Mean rel. error
Vision only	3240	identical	340 (10.5%)	0.029
Fused	3240	identical	**0**	**0.000**

The experiment results at navigation level indicates that the F_1 difference between 0.923 and 1.000 is huger in real application. Although the two maps reflect identical reachability, path length indicators reflect a different conclusion: a planned path shorter than the actual shortest path appears within every ten path queries: under the worst condition, a planned path is 83% shorter than the actual one. The reason is the planner mistakenly formulates a path when there is actually wall partition crossing through. Quadruped robot that navigate through such path will collide with the wall. On the other hand, the fusion map can accurately recover all realistic paths and avoid wrong lengths.

V-B False Alarm Control

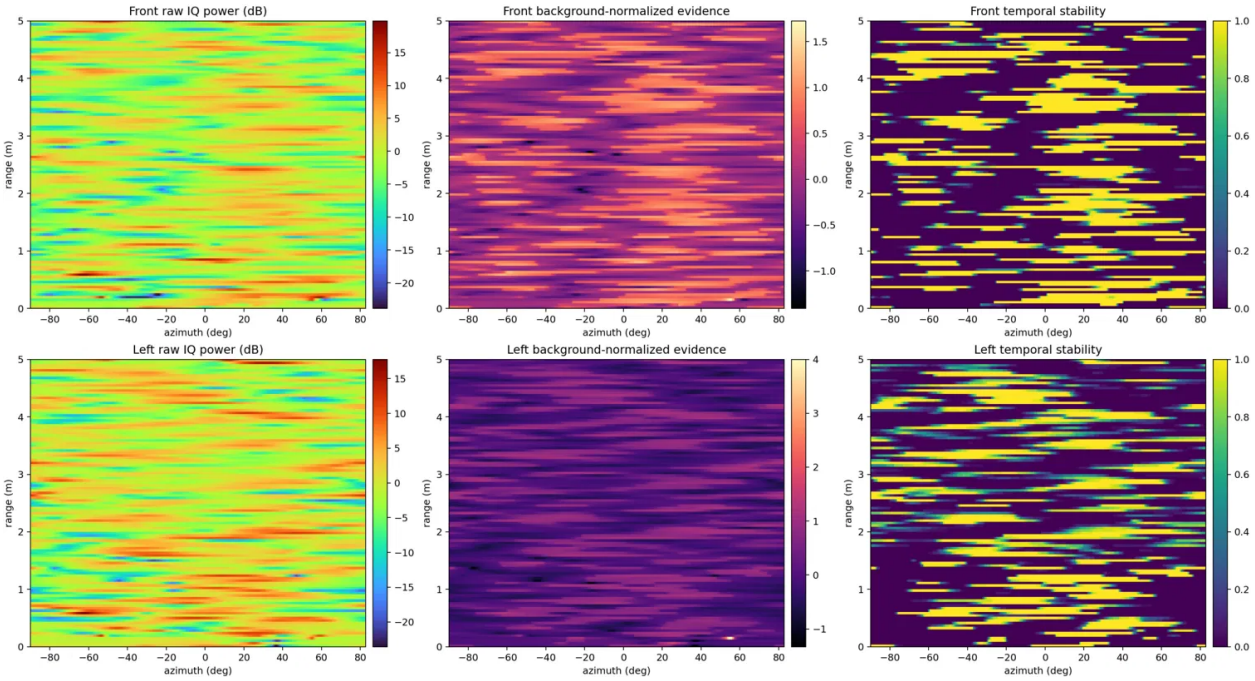


Figure 10: The three Stage B representations for the front view (top) and the left view (bottom).

The leftmost images (Fig. 10) are raw I/Q power maps, banding stripes changing with range but not separate wall signals. The middle images are the results after angular contrast processing, removing many banding stripes and getting localized maximums, while it still cannot be used as map. The rightmost images represent further results after passing through stability gate, with continuous, intense echoes in the form of yellow banding stripes. These are the sequential processes to obtain wall evidence.

The empty scene calibration gives threshold of $\tau = 8.13\,\mathrm{dB}$, which is applied on the stability gate. The image at left is not blank (Fig. 11). Angular excess at the empty room scene can reach 1.3 dB within the ROI. The image at right illustrates the occupancy map, with zero cell occupied, after passing though false alarm control through the empty room threshold, successfully filtering the background noise and false wall evidence. This proves that the radar detection results are real measurement signals instead of ghost targets under scenarios with objects after passing through power intensity gate and stability gate.

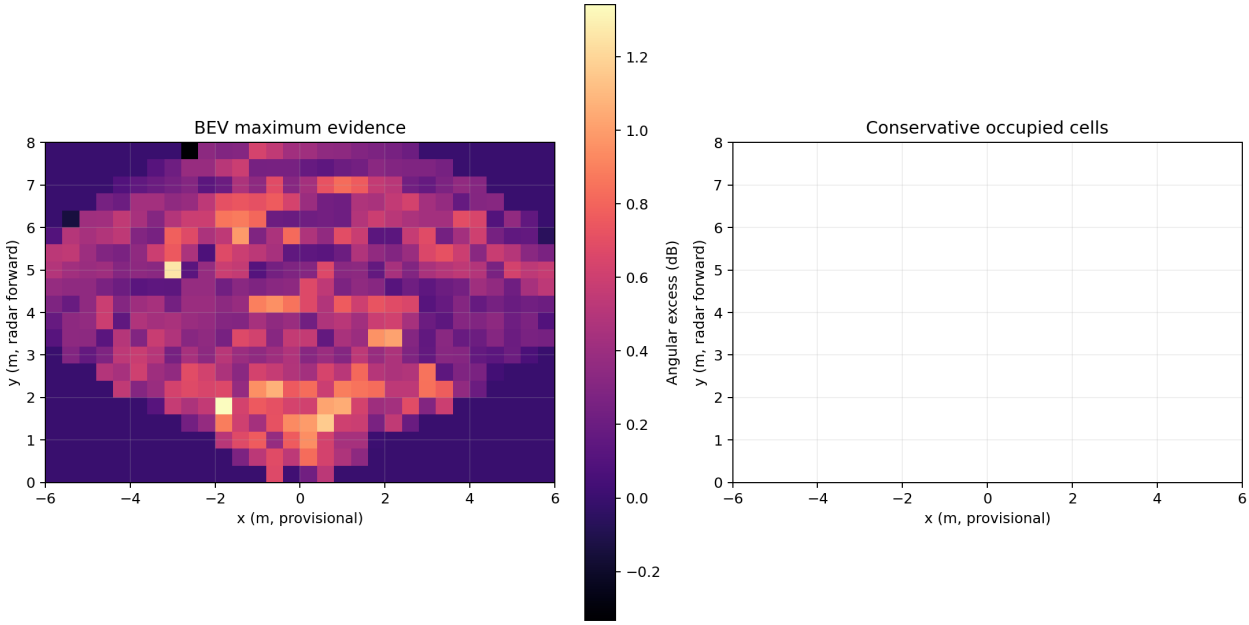


Figure 11: Left: the angular excess on the empty scene BEV occupancy map. Right: none of the cells are identified as wall partition after passing through false alarm control in Eq.(5)

V-C Radar Evidence Support and Regularization

The direct comparison between DA3-only map, fusion map, and ground truth map are illustrated below (Fig. 12), showing better alignment between fusion map and ground truth map.

There is an audit trail for wall edges the radar term saves (Table 6):

Table 6: The chain effect of fused edge score, subsequently saving three edges.

	Edge	DA3	Fused	Restored by
1	H y=2.8, $x\in[2.4, 2.8]$	0.650	0.717	run-length rule
2	H y=2.8, $x\in[2.8, 3.2]$	0.443	0.564	**radar, directly**
3	V x=3.2, $y\in[2.4, 2.8]$	0.104	0.135	endpoint attachment

The three recovered wall edges are not due to independent decisions. Radar evidence only supports the second edge: when its visual score is 0.443, falling within the range Eq.(15) gives, the radar term enables it to exceed the acceptance threshold. The first edge has a visual score or 0.650 exceeding the threshold, but it is an isolated segment with a length of one cell, therefore deleted by the continuation regularization rule. When the second edge is detected, the two edges togther form a segment with length of two cells, saving the first edge. Additionally, the third edge's score is much lower than the threshold, but it is saved due to endpoint attachment regularization. This chain effect demonstrates the gain of regularization.

To further investigate the effectivenss of regularization in stage E (Fig. 13), rules 2, 3 and 4 are deactivated independently. Results show regularization rules have large effect on the F_1 score.

Under all regularization settings, fusion scenarios have better performance, between 0.028 and 0.080, than the vision scenarios. This shows the enhanced performance is not due to man-made regularization. Nonetheless, regularization takes part of many works during reconstruction to enhance F_1 score in both scenarios. When only turning on the boundary assertion rule (rule 1), vision scenario's F_1 score decreases from 0.923 to 0.829, and the fused scenario's F_1 score declines form 1.000 to 0.857.

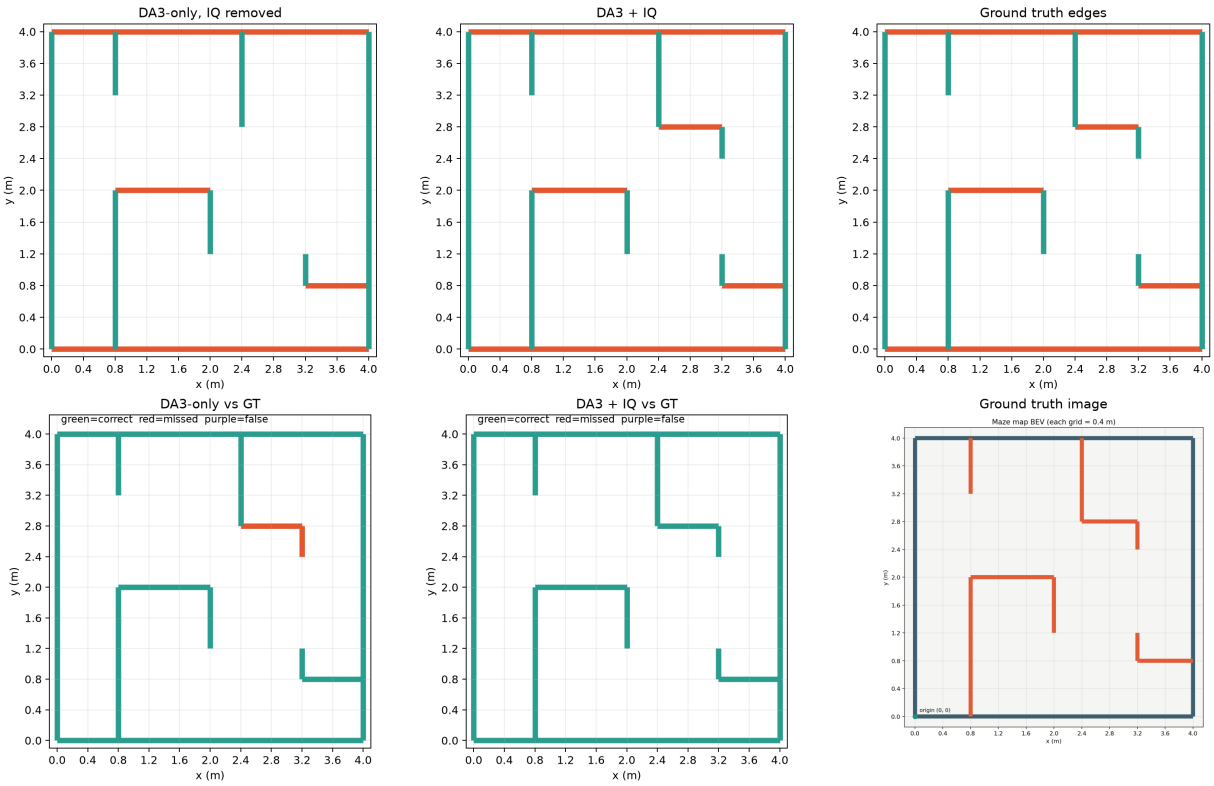


Figure 12: Radar evidence saves three wall edges missed by vision evidence.

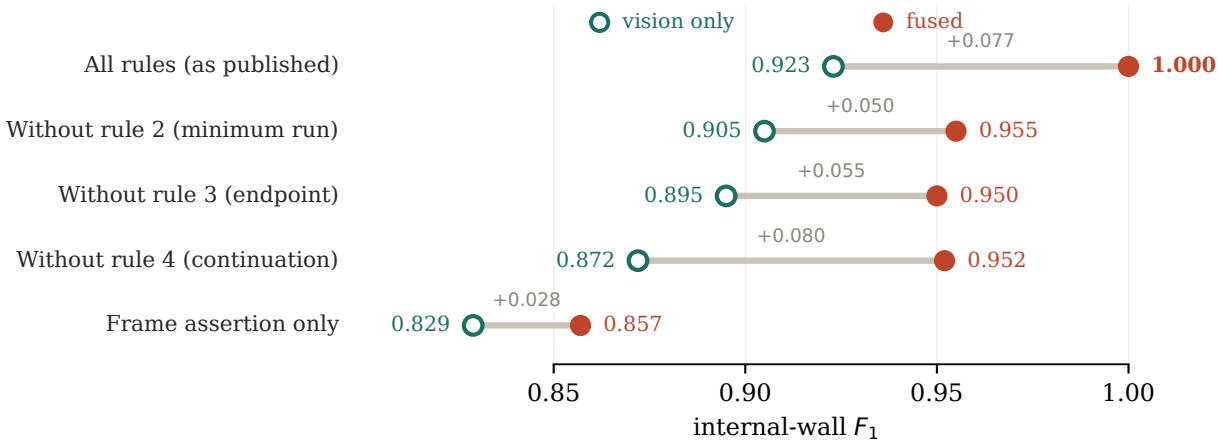


Figure 13: The F_1 of internal walls with Stage E rules partially disabled, and the frame assertion is always active.

V-D Parameter Sensitivity

Parameter sensitivity experiment is conducted on the acceptance threshold η and radar weight ω_i. The fused F_1 maintains 1.000 even as η increases and vision F_1 decreases, and results reflect fused F_1 sustains 1.000 at a larger range.

Table 7: E1. Sensitivity of internal wall F_1 to the acceptance threshold and radar weight.

η	vision	fused	w_i	fused
0.40	1.000	1.000	0.04	1.000
0.44	1.000	1.000	0.08	1.000
0.46	0.923	1.000	**0.10**	**1.000**
0.48	**0.923**	**1.000**	0.14	1.000
0.50	0.923	1.000	0.20	1.000
0.54	0.895	1.000	0.50	0.417
0.58	0.778	0.833	—	—

The parameter experiment verifies the design characteristic of Eq.(15). When the radar weight ω_i is between 0.04 and 0.20 with a factor of five, the performance of the fused outputs remains the same. The reason is the radar term can only recover edges that have scores close to the acceptance threshold. When $\omega_i = 0.50$, which exceeds the acceptance threshold η, the original constraint fails: radar can generate wall edges on itself, with a F_1 score plummeting to 0.417. The threshold experiment then shows performance improvement persists throughout $\eta \in [0.46, 0.54]$ instead of only at a certain single value.

VI Conclusion

This paper presents the RIME system, fusing mmWave radar and monocular visual evidence and executing binary classification on candidate wall edges instead of constructing dense field. The system performs range-azimuth mapping, empty scene threshold setting, stability gating, point cloud operations, and grid edge fusion to produce the final map, having granularity aligning with the planner, that enters the navigation module.

Three core conclusions are obtained. Firstly, the two modalities are both essential. Only using radar evidence, the F_1 is 0.310 at maximum; only using vision evidence, the F_1 is 0.923 at maximum, missing three wall edges. After fusion, the F_1 is improved to 1.000. Secondly, the threshold design can ensure performance, not only attributing the recovered edges to one of the modalities but also preventing mmWave radar, which has limited azimuth resolution, from generating a wall on itself. Thirdly, the improvement produced by fusion has bigger meaning than the difference between F_1 scores on the robotic action level. While 340 planned paths cross through walls not mapped during the 3240 planning queries for the vision map, it does not happen in fused maps.

Future works should include expanding on more indoor layout structures to increase its generalizability, raising mmWave radar resolution for detailed information [10, 29] and better performance, and repeating tests through rooms of smoke and dust [19, 24] to transform this paper's design motive into quantitatively measured results.

References

[1] Independent Committee on the Tai Po Fire, "Final investigation report on the fire at Wang Fuk Court, Tai Po," Government of the Hong Kong Special Administrative Region, Tech. Rep., 2026, fire of 26

November 2025; 168 fatalities; report of 1142 pages released August 2026.

[2] National Institute for Occupational Safety and Health, "Nine career fire fighters die in rapid fire progression at commercial furniture showroom—South Carolina," NIOSH Fire Fighter Fatality Investigation and Prevention Program, Tech. Rep. Report F2007-18, 2009, incident of 18 June 2007; report released 11 February 2009.

[3] W. R. Mora, "Understanding and solving firefighter disorientation," *Fire Engineering*, 2003, review of NIOSH line-of-duty-death investigations; 17 disorientation cases accounting for 23 fatalities.

[4] A. Radford, J. W. Kim, C. Hallacy, A. Ramesh, G. Goh, S. Agarwal, G. Sastry, A. Askell, P. Mishkin, J. Clark, G. Krueger, and I. Sutskever, "Learning transferable visual models from natural language supervision," in *International Conference on Machine Learning (ICML)*, 2021, pp. 8748–8763.

[5] M. Savva, A. Kadian, O. Maksymets, Y. Zhao, E. Wijmans, B. Jain, J. Straub, J. Liu, V. Koltun, J. Malik, D. Parikh, and D. Batra, "Habitat: A platform for embodied AI research," in *Proceedings of the IEEE/CVF International Conference on Computer Vision (ICCV)*, 2019, pp. 9339–9347.

[6] K. Burnett, Y. Wu, D. J. Yoon, A. P. Schoellig, and T. D. Barfoot, "Are we ready for radar to replace lidar in all-weather mapping and localization?" *IEEE Robotics and Automation Letters*, vol. 7, no. 4, pp. 10 328–10 335, 2022.

[7] K. Harlow, H. Jang, T. D. Barfoot, A. Kim, and C. Heckman, "A new wave in robotics: Survey on recent mmWave radar applications in robotics," *IEEE Transactions on Robotics*, vol. 40, pp. 4544–4560, 2024.

[8] A. Venon, Y. Dupuis, P. Vasseur, and P. Merriaux, "Millimeter wave FMCW radars for perception, recognition and localization in automotive applications: A survey," *IEEE Transactions on Intelligent Vehicles*, vol. 7, no. 3, pp. 533–555, 2022.

[9] H. Kong, C. Huang, J. Yu, and X. Shen, "A survey of mmWave radar-based sensing in autonomous vehicles, smart homes and industry," *IEEE Communications Surveys & Tutorials*, 2024.

[10] Y.-J. Li, S. Hunt, J. Park, M. O'Toole, and K. Kitani, "Azimuth super-resolution for FMCW radar in autonomous driving," in *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*, 2023, pp. 17 504–17 513.

[11] H. Lin, S. Chen, J. H. Liew, D. Y. Chen, Z. Li, G. Shi, J. Feng, and B. Kang, "Depth anything 3: Recovering the visual space from any views," *arXiv preprint arXiv:2511.10647*, 2025.

[12] L. Yang, B. Kang, Z. Huang, Z. Zhao, X. Xu, J. Feng, and H. Zhao, "Depth anything v2," *Advances in Neural Information Processing Systems*, vol. 37, pp. 21 875–21 911, 2024.

[13] H. Rohling, "Radar CFAR thresholding in clutter and multiple target situations," *IEEE Transactions on Aerospace and Electronic Systems*, vol. AES-19, no. 4, pp. 608–621, 1983.

[14] M. I. Skolnik, *Radar Handbook*, 3rd ed. McGraw-Hill Education, 2008.

[15] Y. Zhou, L. Liu, H. Zhao, M. López-Benítez, L. Yu, and Y. Yue, "Towards deep radar perception for autonomous driving: Datasets, methods, and challenges," *Sensors*, vol. 22, no. 11, p. 4208, 2022.

[16] Z. Hong, Y. Petillot, and S. Wang, "RadarSLAM: Radar based large-scale SLAM in all weathers," in *IEEE/RSJ International Conference on Intelligent Robots and Systems (IROS)*, 2020, pp. 5164–5170.

[17] D. Adolfsson, M. Magnusson, A. Alhashimi, A. J. Lilienthal, and H. Andreasson, "Lidar-level localization with radar? the CFEAR approach to accurate, fast, and robust large-scale radar odometry in diverse environments," *IEEE Transactions on Robotics*, vol. 39, no. 2, pp. 1476–1495, 2023.

[18] Y. S. Park, Y.-S. Shin, J. Kim, and A. Kim, "3D ego-motion estimation using low-cost mmWave radars via radar velocity factor for pose-graph SLAM," *IEEE Robotics and Automation Letters*, vol. 6, no. 4, pp. 7691–7698, 2021.

[19] A. Kramer, C. Stahoviak, A. Santamaria-Navarro, A.-a. Agha-Mohammadi, and C. Heckman, "Radar-inertial ego-velocity estimation for visually degraded environments," in *IEEE International Conference on Robotics and Automation (ICRA)*, 2020, pp. 5739–5746.

[20] Y. Almalioglu, M. Turan, C. X. Lu, N. Trigoni, and A. Markham, "Milli-RIO: Ego-motion estimation with low-cost millimetre-wave radar," *IEEE Sensors Journal*, vol. 21, no. 3, 2021.

[21] N. J. Abu-Alrub and N. A. Rawashdeh, "Radar odometry for autonomous ground vehicles: A survey of methods and datasets," *IEEE Transactions on Intelligent Vehicles*, vol. 9, no. 3, pp. 4275–4291, 2024.

[22] A. Kramer, K. Harlow, C. Williams, and C. Heckman, "ColoRadar: The direct 3d millimeter wave radar dataset," *The International Journal of Robotics Research*, vol. 41, no. 4, pp. 351–360, 2022.

[23] D. Barnes, M. Gadd, P. Murcutt, P. Newman, and I. Posner, "The Oxford radar RobotCar dataset: A radar extension to the Oxford RobotCar dataset," in *IEEE International Conference on Robotics and Automation (ICRA)*, 2020, pp. 6433–6438.

[24] C. X. Lu, S. Rosa, P. Zhao, B. Wang, C. Chen, J. A. Stankovic, N. Trigoni, and A. Markham, "See through smoke: Robust indoor mapping with low-cost mmWave radar," in *Proceedings of the 18th International Conference on Mobile Systems, Applications, and Services (MobiSys)*, 2020, pp. 14–27.

[25] M. Aladsani, A. Alkhateeb, and G. C. Trichopoulos, "Leveraging mmWave imaging and communications for simultaneous localization and mapping," in *IEEE International Conference on Acoustics, Speech and Signal Processing (ICASSP)*, 2019, pp. 4539–4543.

[26] A. Yassin, Y. Nasser, A. Y. Al-Dubai, and M. Awad, "MOSAIC: Simultaneous localization and environment mapping using mmWave without a-priori knowledge," *IEEE Access*, vol. 6, pp. 68 932–68 947, 2018.

[27] J. Guan, S. Madani, S. Jog, S. Gupta, and H. Hassanieh, "Through fog high-resolution imaging using millimeter wave radar," in *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*, 2020, pp. 11 464–11 473.

[28] C. X. Lu, M. R. U. Saputra, P. Zhao, Y. Almalioglu, P. P. B. de Gusmao, C. Chen, K. Sun, N. Trigoni, and A. Markham, "milliEgo: Single-chip mmWave radar aided egomotion estimation via deep sensor fusion," in *Proceedings of the 18th ACM Conference on Embedded Networked Sensor Systems (SenSys)*, 2020, pp. 109–122.

[29] E. Sie, Z. Liu, and D. Vasisht, "BatMobility: Towards flying without seeing for autonomous drones," in *Proceedings of the 29th Annual International Conference on Mobile Computing and Networking (MobiCom)*, 2023, pp. 1–16.

[30] S. Wang, V. Leroy, Y. Cabon, B. Chidlovskii, and J. Revaud, "DUSt3R: Geometric 3D vision made easy," in *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*, 2024, pp. 20 697–20 709.

[31] V. Leroy, Y. Cabon, and J. Revaud, "Grounding image matching in 3D with MASt3R," in *European Conference on Computer Vision (ECCV)*, 2024, pp. 71–91.

[32] J. Wang, M. Chen, N. Karaev, A. Vedaldi, C. Rupprecht, and D. Novotny, "VGGT: Visual geometry grounded transformer," in *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*, 2025, pp. 5294–5306.

[33] L. Yang, B. Kang, Z. Huang, X. Xu, J. Feng, and H. Zhao, "Depth anything: Unleashing the power of large-scale unlabeled data," in *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*, 2024, pp. 10 371–10 381.

[34] H. Lin, S. Peng, J. Chen, S. Peng, J. Sun, M. Liu, H. Bao, J. Feng, X. Zhou, and B. Kang, "Prompting depth anything for 4K resolution accurate metric depth estimation," in *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*, 2025, pp. 17 070–17 080.

[35] R. Murai, E. Dexheimer, and A. J. Davison, "MASt3R-SLAM: Real-time dense SLAM with 3D reconstruction priors," in *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*, 2025.

[36] D. Maggio, H. Lim, and L. Carlone, "VGGT-SLAM: Dense RGB SLAM optimized on the SL(4) manifold," *Advances in Neural Information Processing Systems*, 2025.

[37] H. Wang and L. Agapito, "3D reconstruction with spatial memory," in *International Conference on 3D Vision (3DV)*, 2025, pp. 78–89.

[38] R. Nabati and H. Qi, "CenterFusion: Center-based radar and camera fusion for 3D object detection," in *Proceedings of the IEEE/CVF Winter Conference on Applications of Computer Vision (WACV)*, 2021, pp. 1527–1536.

[39] J.-J. Hwang, H. Kretzschmar, J. Manela, S. Rafferty, N. Armstrong-Crews, T. Chen, and D. Anguelov, "CramNet: Camera-radar fusion with ray-constrained cross-attention for robust 3D object detection," in *European Conference on Computer Vision (ECCV)*, 2022, pp. 388–405.

[40] Y. Kim, J. Shin, S. Kim, I.-J. Lee, J. W. Choi, and D. Kum, "CRN: Camera radar net for accurate, robust, efficient 3D perception," in *Proceedings of the IEEE/CVF International Conference on Computer Vision (ICCV)*, 2023.

[41] Z. Lin, Z. Liu, Z. Xia, X. Wang, Y. Wang, S. Qi, Y. Dong, N. Dong, L. Zhang, and C. Zhu, "RCBEVDet: Radar-camera fusion in bird's eye view for 3D object detection," in *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*, 2024.

[42] Z. Han, J. Wang, Z. Xu, S. Yang, L. He, S. Xu, and J. Wang, "4D millimeter-wave radar in autonomous driving: A survey," *arXiv preprint arXiv:2306.04242*, 2023.

[43] H. Caesar, V. Bankiti, A. H. Lang, S. Vora, V. E. Liong, Q. Xu, A. Krishnan, Y. Pan, G. Baldan, and O. Beijbom, "nuScenes: A multimodal dataset for autonomous driving," in *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*, 2020.

[44] A. Elfes, "Using occupancy grids for mobile robot perception and navigation," *Computer*, vol. 22, no. 6, pp. 46–57, 1989.

[45] C. Cadena, L. Carlone, H. Carrillo, Y. Latif, D. Scaramuzza, J. Neira, I. Reid, and J. J. Leonard, "Past, present, and future of simultaneous localization and mapping: Toward the robust-perception age," *IEEE Transactions on Robotics*, vol. 32, no. 6, pp. 1309–1332, 2016.

[46] W. Hess, D. Kohler, H. Rapp, and D. Andor, "Real-time loop closure in 2D LIDAR SLAM," in *IEEE International Conference on Robotics and Automation (ICRA)*, 2016, pp. 1271–1278.

[47] C. Campos, R. Elvira, J. J. G. Rodríguez, J. M. M. Montiel, and J. D. Tardós, "ORB-SLAM3: An accurate open-source library for visual, visual-inertial, and multimap SLAM," *IEEE Transactions on Robotics*, vol. 37, no. 6, pp. 1874–1890, 2021.

[48] M. A. Fischler and R. C. Bolles, "Random sample consensus: A paradigm for model fitting with applications to image analysis and automated cartography," *Communications of the ACM*, vol. 24, no. 6, pp. 381–395, 1981.

[49] R. O. Duda and P. E. Hart, "Use of the Hough transformation to detect lines and curves in pictures," *Communications of the ACM*, vol. 15, no. 1, pp. 11–15, 1972.

[50] J. M. Coughlan and A. L. Yuille, "Manhattan world: Compass direction from a single image by Bayesian inference," in *Proceedings of the IEEE International Conference on Computer Vision (ICCV)*, 1999.

[51] P. E. Hart, N. J. Nilsson, and B. Raphael, "A formal basis for the heuristic determination of minimum cost paths," *IEEE Transactions on Systems Science and Cybernetics*, vol. 4, no. 2, pp. 100–107, 1968.

[52] B. Yamauchi, "A frontier-based approach for autonomous exploration," in *Proceedings of the IEEE International Symposium on Computational Intelligence in Robotics and Automation (CIRA)*, 1997, pp. 146–151.

[53] ——, "Frontier-based exploration using multiple robots," in *Proceedings of the Second International Conference on Autonomous Agents*, 1998, pp. 47–53.

[54] W. Burgard, M. Moors, C. Stachniss, and F. E. Schneider, "Coordinated multi-robot exploration," *IEEE Transactions on Robotics*, vol. 21, no. 3, pp. 376–386, 2005.

[55] B. Zhou, Y. Zhang, X. Chen, and S. Shen, "FUEL: Fast UAV exploration using incremental frontier structure and hierarchical planning," *IEEE Robotics and Automation Letters*, vol. 6, no. 2, pp. 779–786, 2021.

[56] A. Bircher, M. Kamel, K. Alexis, H. Oleynikova, and R. Siegwart, "Receding horizon next-best-view planner for 3D exploration," in *IEEE International Conference on Robotics and Automation (ICRA)*, 2016, pp. 1462–1468.

[57] S. Bai, K. Chen, X. Liu, J. Wang, W. Ge, S. Song, K. Dang, P. Wang, S. Wang, J. Tang *et al.*, "Qwen2.5-VL technical report," *arXiv preprint arXiv:2502.13923*, 2025.

[58] S. Y. Gadre, M. Wortsman, G. Ilharco, L. Schmidt, and S. Song, "CLIP on wheels: Zero-shot object navigation as object localization and exploration," *arXiv preprint arXiv:2203.10421*, 2022.

[59] D. Shah, B. Osiński, B. Ichter, and S. Levine, "LM-Nav: Robotic navigation with large pre-trained models of language, vision, and action," in *Conference on Robot Learning (CoRL)*, 2023, pp. 492–504.

[60] C. Huang, O. Mees, A. Zeng, and W. Burgard, "Visual language maps for robot navigation," in *IEEE International Conference on Robotics and Automation (ICRA)*, 2023, pp. 10 608–10 615.

[61] B. Yu, H. Kasaei, and M. Cao, "L3MVN: Leveraging large language models for visual target navigation," in *IEEE/RSJ International Conference on Intelligent Robots and Systems (IROS)*, 2023, pp. 3554–3560.

[62] K. Zhou, K. Zheng, C. Pryor, Y. Shen, H. Jin, L. Getoor, and X. E. Wang, "ESC: Exploration with soft commonsense constraints for zero-shot object navigation," in *International Conference on Machine Learning (ICML)*, 2023, pp. 42 829–42 842.

[63] N. Yokoyama, S. Ha, D. Batra, J. Wang, and B. Bucher, "VLFM: Vision-language frontier maps for zero-shot semantic navigation," in *IEEE International Conference on Robotics and Automation (ICRA)*, 2024, pp. 42–48.

[64] S. K. Ramakrishnan, A. Gokaslan, E. Wijmans, O. Maksymets, A. Clegg, J. M. Turner, E. Undersander, W. Galuba, A. Westbury, A. X. Chang, M. Savva, Y. Zhao, and D. Batra, "Habitat-matterport 3D dataset (HM3D): 1000 large-scale 3D environments for embodied AI," in *Thirty-fifth Conference on Neural Information Processing Systems Datasets and Benchmarks Track*, 2021.

[65] B. Kehoe, S. Patil, P. Abbeel, and K. Goldberg, "A survey of research on cloud robotics and automation," *IEEE Transactions on Automation Science and Engineering*, vol. 12, no. 2, pp. 398–409, 2015.

[66] W. Shi, J. Cao, Q. Zhang, Y. Li, and L. Xu, "Edge computing: Vision and challenges," *IEEE Internet of Things Journal*, vol. 3, no. 5, pp. 637–646, 2016.